Business Understanding

Introduction:

Business understanding is the first and most important step in any data science or machine learning project. It focuses on clearly identifying the business problem, defining objectives, and understanding how data can help solve real-world challenges. Before performing data analysis or building models, it is necessary to understand the domain, goals, stakeholders, and expected outcomes.

Selected Domain: Online Food Delivery Service

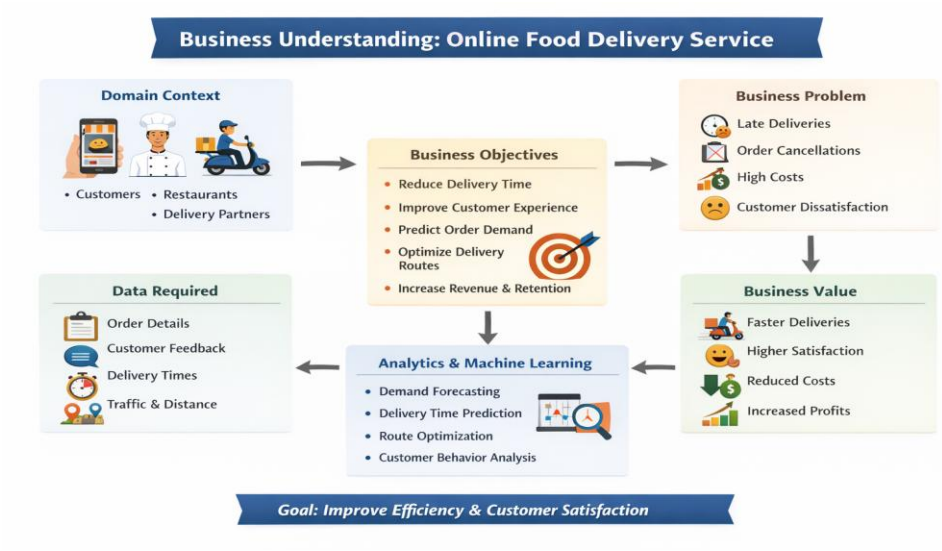
The selected domain is Online Food Delivery, where digital platforms connect customers, restaurants, and delivery partners. These services allow users to order food online and receive it at their location quickly and conveniently.

Business Problem

Online food delivery companies face several operational challenges such as:

- Late deliveries
- Order cancellations
- Customer dissatisfaction
- Inefficient delivery routes
- High operational and fuel costs

The business aims to improve efficiency while maintaining customer satisfaction.



Business Objectives:

The major objectives include:

- Reduce average delivery time
- Improve customer experience and ratings
- Predict order demand during peak hours
- Optimize delivery partner allocation
- Increase revenue and customer retention

Key Stakeholders:

- Customers – expect fast and accurate delivery
- Restaurants – want more orders and visibility
- Delivery Partners – require optimized routes and fair workload
- Company Management – focuses on profit and growth

Business Process Overview

The food delivery workflow includes:

1. Customer places an order through the application.
2. Restaurant accepts and prepares the order.
3. Delivery partner is assigned automatically.
4. Order is picked up and delivered.
5. Customer provides rating and feedback.

Each stage produces valuable data useful for analysis and improvement.

Business Questions

Important questions the business wants to answer:

- When do most orders occur?
- Which areas have the highest demand?
- Why do deliveries get delayed?
- Which customers are likely to cancel orders?

Data Required

To address these questions, the business collects:

- Order details (time, location, price)
- Customer ratings and feedback
- Delivery time records
- Restaurant preparation time
- Distance and traffic information

Challenges Faced by the Business

- Traffic congestion affecting delivery speed
- Poor demand prediction during peak hours
- Uneven distribution of delivery partners
- Restaurant preparation delays
- Customer complaints due to late delivery

Role of Data Analytics

Data analytics helps by:

- Predicting busy hours using historical data
- Identifying high-demand regions
- Improving delivery route planning
- Understanding customer behavior
- Reducing operational costs

Possible Machine Learning Solutions

Machine learning techniques can be used to:

- Predict delivery time accurately
- Recommend restaurants to users
- Forecast future demand
- Detect cancellation probability
- Optimize delivery partner assignment

Success Criteria

The project is considered successful if:

- Delivery time decreases
- Customer ratings increase
- Order cancellations reduce
- Operational efficiency improves

Business Value Generated

Applying data-driven solutions helps achieve:

- Faster delivery services
- Better customer satisfaction
- Increased profits
- Efficient resource utilization
- Strong competitive advantage

Conclusion

Business understanding connects technical solutions with real-world business needs. In the online food delivery domain, identifying operational challenges and analyzing customer behavior enables companies to make informed decisions. A clear understanding of business goals ensures that data science solutions create measurable business value and improve overall service performance.